Exploratory Data Analysis and Linear Regression on a House Price Dataset

Understanding housing price patterns and modelling the link between square footage and price

Project Title	Exploratory Data Analysis and Linear Regression on a House Price Dataset
Student Name	P Rehan Khan
USN / Batch	1BF25AI099
Institution	BMS College of Engineering
Date	October 25, 2026
Dataset Source	https://www.kaggle.com/c/house-prices-advanced-regression-techniques
GitHub Repository	https://github.com/Rehan64-source/Rehan_khan_House_Price_EDA_Project

2. Objective

The goal of this project is to perform Exploratory Data Analysis (EDA) to understand housing price patterns, treat outliers, and build a linear regression model to analyse the direct relationship between square footage and house price. The workflow covers data inspection, cleaning (duplicates, missing values and outliers), visual exploration and evaluation of the model on unseen test data.

3. Dataset Overview

After removing 10 duplicate rows, the final dataset has **1,000 rows and 8 columns** (the raw file had 1,010 rows). All eight columns are numerical. The **target variable** is **House_Price**, the price of the house, which the regression model predicts from **Square_Footage**.

Numerical columns

Column	Dtype	Description
Square_Footage	float64	Living area of the house in square feet
Num_Bedrooms	int64	Number of bedrooms
Num_Bathrooms	int64	Number of bathrooms
Year_Built	int64	Year the house was built
Lot_Size	float64	Size of the plot
Garage_Size	int64	Garage size (number of cars / spaces)
Neighborhood_Quality	int64	Neighbourhood quality rating (ordinal score)
House_Price	float64	**Target variable:** selling price of the house

Dataset size before and after cleaning

Stage	Rows	Columns	Missing values	Duplicates
Raw data	1,010	8	10 (5 Square_Footage + 5 House_Price)	10
After cleaning	1,000	8	0	0

4. Data Quality and Cleaning

The following checks and decisions were applied in the notebook:

Step	Result and decision
Inspect structure	Data loaded and inspected with head(), tail() and info(): 1,010 rows x 8 columns; 3 float64 and 5 int64 columns, so no type conversion was needed.
Duplicate rows	**10 duplicate rows** were found and removed, leaving 1,000 rows.
Missing values	**5 missing values in Square_Footage** and **5 in House_Price**. Both were filled with the **mean** of the column so that all rows are kept and data integrity is preserved. A re-check confirmed 0 missing values.
Outlier detection	Box plots of House_Price and Square_Footage showed extreme values (Chart 2): five very high prices between about 2.5 and 5 million, and a few very large houses of about 9,000 to 18,000 sq ft (Chart 1).
Outlier treatment	Outliers in **both the price and footage columns were capped using the IQR method**: values above Q3 + 1.5 x IQR were set to the upper bound and values below Q1 - 1.5 x IQR to the lower bound. Capping keeps every row while limiting the influence of extreme values (Charts 3 and 4).

5. Descriptive Analysis

Output of **df.head()** (first 5 rows):

Square_ Footage	Num_ Bedrooms	Num_ Bathrooms	Year_ Built	Lot_ Size	Garage_ Size	Neighborhood_ Quality	House_ Price
1360	2	1	1981	0.60	0	5	262,383
4272	3	3	2016	4.75	1	6	985,261
3592	1	2	2016	3.63	0	9	777,977
966	1	2	1977	2.73	1	8	229,699
4926	2	1	1993	4.70	0	8	1,041,741

Output of **df.info()** on the raw data:

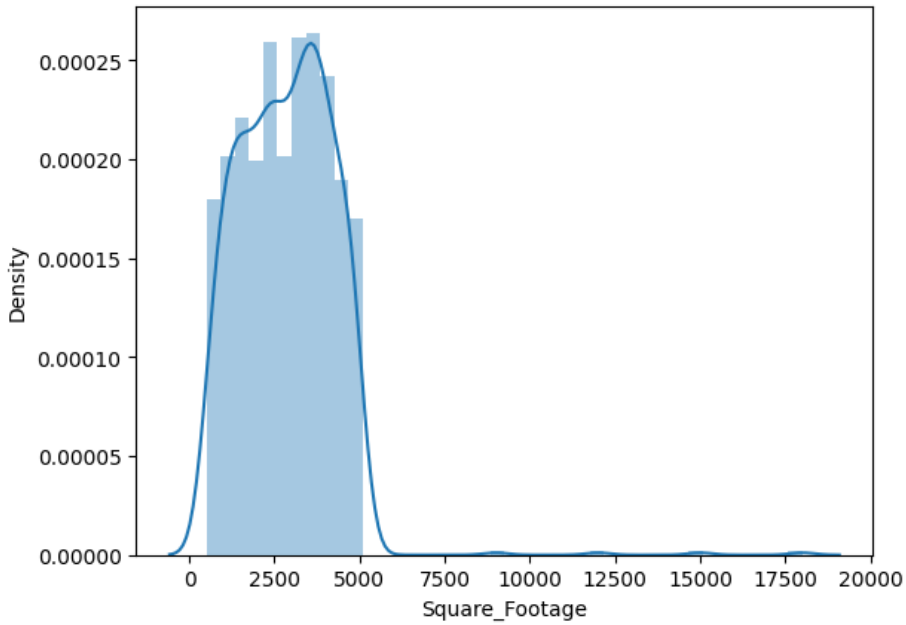
Column	Non-null count	Dtype
Square_Footage	1005 non-null (5 missing)	float64
Num_Bedrooms	1010 non-null	int64
Num_Bathrooms	1010 non-null	int64
Year_Built	1010 non-null	int64
Lot_Size	1010 non-null	float64
Garage_Size	1010 non-null	int64
Neighborhood_Quality	1010 non-null	int64
House_Price	1005 non-null (5 missing)	float64

Observation: all eight columns are numeric (3 float64, 5 int64), so no encoding is needed. Only Square_Footage and House_Price contained missing values (5 each); every other column was complete. House prices in the sample rows range from about 230,000 to over 1,000,000.

6. Visual EDA

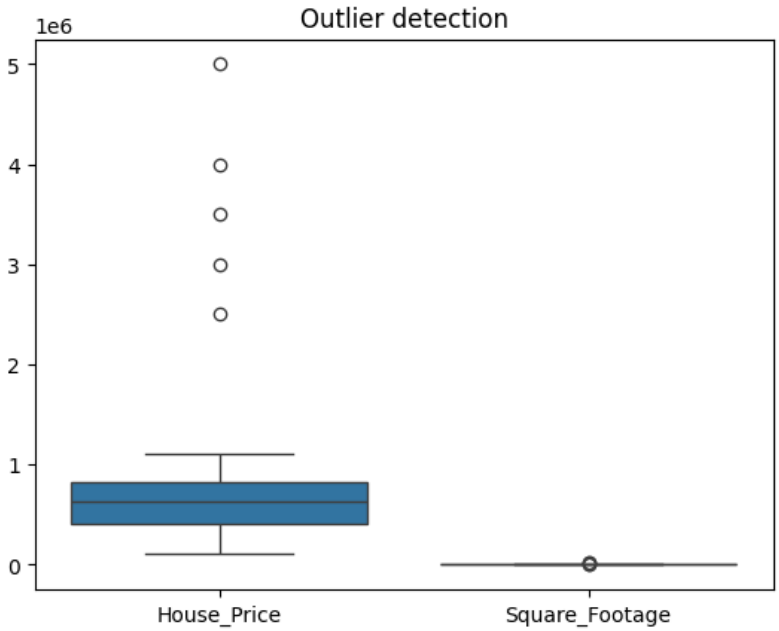
Six charts are shown below: density histograms, box plots and regression (scatter) plots. Each has a title and an observation.

Chart 1: Distribution of Square_Footage (before outlier capping)



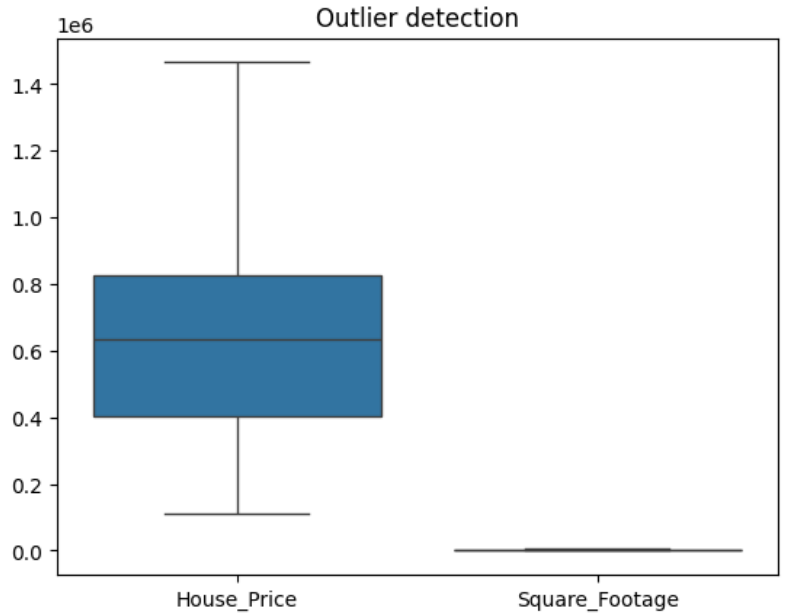
Observation: Most houses are between about 500 and 5,000 sq ft, with the density highest around 2,000 to 4,000 sq ft. A few extreme houses of roughly 9,000 to 18,000 sq ft form a long, thin tail, which stretches the axis to 20,000 and flags outliers.

Chart 2: Box Plot for Outlier Detection, House_Price and Square_Footage (before capping)



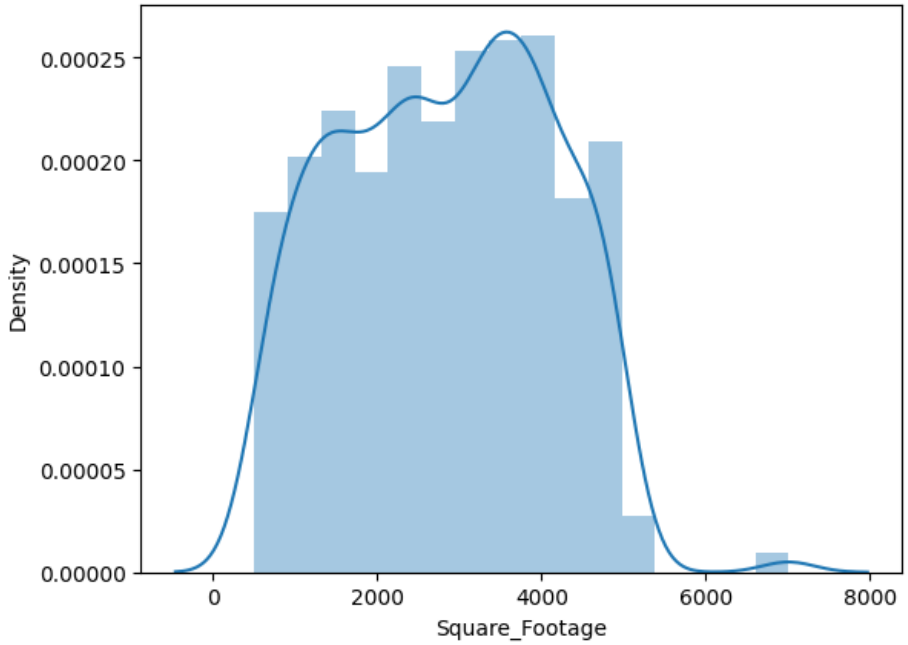
Observation: Five House_Price points lie far above the upper whisker, between about 2.5 and 5 million, so they are clear outliers. Square_Footage looks flat because it is on the same axis as price (a scale of millions); its outliers are better seen in Chart 1.

Chart 3: Box Plot after IQR Capping, House_Price and Square_Footage



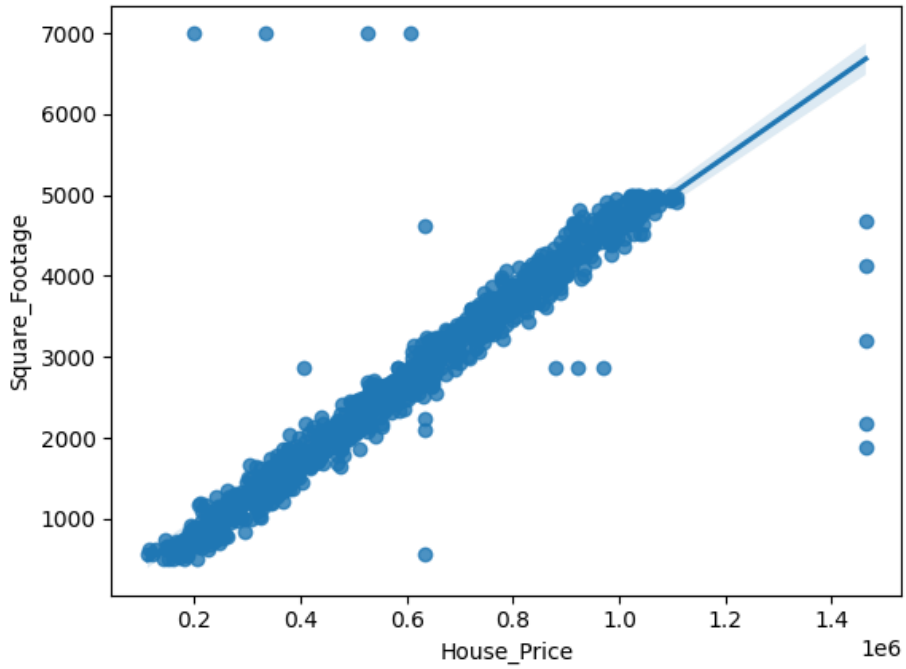
Observation: After capping, no points lie beyond the whiskers. The extreme prices were pulled down to the upper IQR bound of about 1.47 million, and the price box (median about 0.63 million) is now far more readable. This shows the IQR method kept every row but limited extreme values.

Chart 4: Distribution of Square_Footage (after IQR capping)



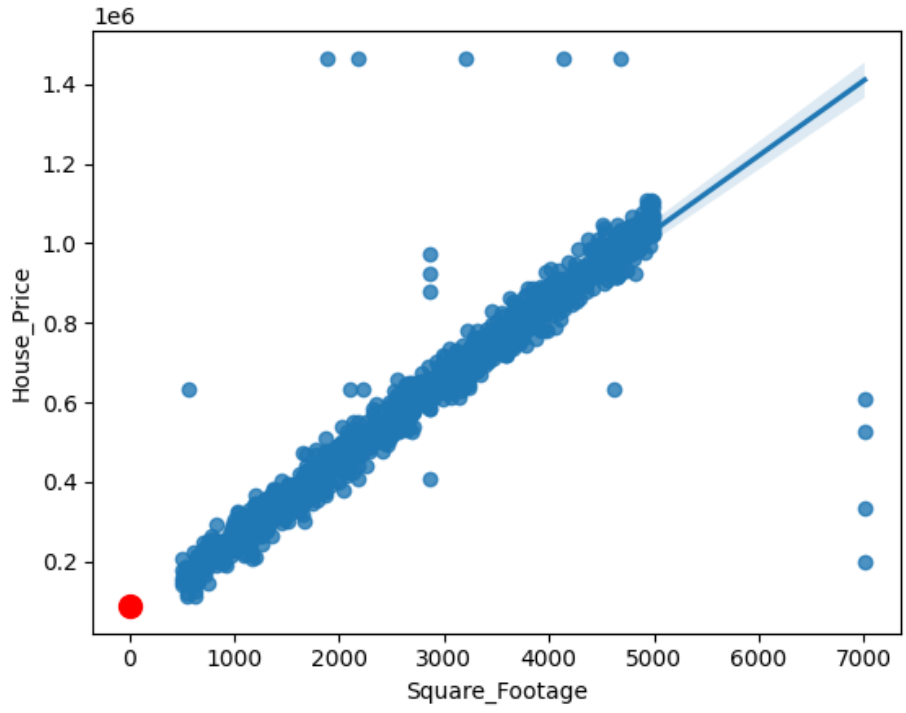
Observation: The distribution now spans about 500 to 5,000 sq ft, with a wide, fairly even spread and a peak near 3,500 sq ft. The extreme houses were capped at about 7,000 sq ft, which appears as a small bump at the right edge.

Chart 5: Regression Plot of Square_Footage against House_Price (price on x-axis)



Observation: The points form a tight, upward straight band: larger houses sell for more, which is a strong positive linear relationship. The few points away from the band at the top and right edges are the capped outliers.

Chart 6: Regression Plot of Square Footage vs House Price with Red Prediction Point



Observation: The fitted line follows the dense band of points closely, supporting a simple linear model. The red point is the model's prediction for the new house (Square_Footage = 6, predicted price about 88,430). It sits far to the left of the data, so it is an extrapolation and not a realistic price.

Linear regression model results

A simple linear regression was trained with **Square_Footage** as the only feature and **House_Price** as the target, using an 80/20 train-test split (800 training rows, 200 test rows, random_state = 42) on the cleaned data.

Metric (on unseen test data)	Value
R-squared score	0.8907 (about 0.89)
Mean Squared Error	7,048,526,538 (about 7.05 x 10^9)
Root Mean Squared Error	about 83,956

Observation: an R-squared of about 0.89 means square footage alone explains roughly 89% of the variation in house price on test data the model had not seen. In the printed predictions the predicted and actual prices are close (for example, 901,000 actual vs 848,000 predicted).

7. Key Findings

1. **Square footage has a strong positive relationship with price.** The regression plots show a tight upward band, so larger houses sell for more (Charts 5 and 6).
2. **The linear regression model performs strongly.** It reached an R-squared of 0.89 on 200 unseen test rows, with an MSE of about 7.05 x 10^9 (RMSE about 83,956).
3. **Missing data was handled successfully.** 10 duplicates were removed and the 5 + 5 missing values were filled with the mean, leaving 1,000 complete rows and 0 missing values.
4. **Extreme values were treated with the IQR method.** Five very high prices (2.5 to 5 million) and a few very large houses (9,000 to 18,000 sq ft) were capped rather than deleted, producing clean box plots (Charts 2 and 3).
5. **House sizes are spread widely.** Most houses are 500 to 5,000 sq ft, with the highest density around 2,000 to 4,000 sq ft (Charts 1 and 4).

Limitations

- The dataset lacks exact geographic location data (like ZIP code or city), which heavily impacts real-world real estate pricing.
- The model uses **only Square_Footage**; other columns such as Year_Built, Neighborhood_Quality and Lot_Size were not used, so predictions ignore them.
- Capping outliers and filling the target (House_Price) with the mean slightly alter the data; capped values appear as stray points in Charts 5 and 6.
- The demonstration prediction used 6 sq ft, far outside the observed range (about 500 to 7,000 sq ft), so it shows the workflow, not a realistic price.
- Results come from a single 80/20 split, and the analysis shows association, not cause.

8. Conclusion

This project cleaned a 1,010-row house price dataset (removing 10 duplicates, filling 5 + 5 missing values with the mean and capping outliers with the IQR method) and explored it with density plots, box plots and regression plots. The analysis shows that square footage has a strong, nearly linear relationship with house price, and a simple linear regression model achieved an R-squared of 0.89, successfully demonstrating predictive capability on unseen test data.

Challenges faced: imputing missing values in both the feature and the target, and calculating the IQR boundaries correctly so that outliers were capped without losing rows.

Future analysis: build a multiple regression model that also uses Year_Built, Neighborhood_Quality, Lot_Size and bedrooms; add location data; try a log transformation of price or other models; and validate the results with cross-validation.